Contact

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Work Experience

July 2025–Present: **Associate Teaching Professor**. Department of Electrical and Computer Engineering, College of Engineering, Northeastern University at Boston, USA.

- I teach both undergraduate and graduate courses in software engineering, machine learning, and large language models.
- I developed and launched a new course on Large Language Models (EECE 5668)—the first of its kind at the university—providing students with an in-depth exploration of the theoretical foundations, training techniques, and practical applications of large language models across a wide range of domains. The course quickly became one of the most popular electives in the department, consistently attracting high enrollment and receiving outstanding student feedback.
- I am authoring a comprehensive three-volume textbook titled Machine Learning Foundations, under contract with Addison-Wesley (Pearson), covering supervised learning, unsupervised learning, deep learning, and advanced topics. Volume I is scheduled for publication in January 2026, with early access already available on O'Reilly Media.
- I am advising a PhD student conducting research on multi-robot systems, with a focus on developing a unified framework for task and path planning in autonomous mining robots.

August 2020–July 2025: **Assistant Teaching Professor**. Department of Electrical and Computer Engineering, College of Engineering, Northeastern University at Boston, USA.

In this role, I was responsible for traditional instruction, curriculum development, and student advising. My primary teaching duties centered on the Master's program in Data Science, including courses in algorithm design, data processing, machine learning, data mining, and data visualization.

November 2018–August 2020: **Postdoctoral Research Associate**. Khoury College of Computer Sciences, Northeastern University at Boston, USA. Hosted by Prof. Christopher Amato.

I have developed a fully decentralized multi-agent learning method based on policy gradients, that tackles the problem of multi-target search and detection by a team of drones.

July 2017–March 2018: **Senior Machine Learning and NLP Research Engineer**, Basis Technology. The tasks in this position included: (a) Keeping up-to-date on academic literature in deep learning, machine learning and NLP, (b) Introducing techniques and ideas to push the envelope on accuracy and performance of the team's existing products, e.g., sentiment analysis, text categorization, and task-specific word embeddings, and (c) Writing code with the team to build new capabilities (e.g., concepts extraction).

October 2013–October 2018: **PhD Student and Teaching Assistant**, Bar-Ilan University, Israel.

I was a PhD student in the robotics lab at Bar-Ilan University, headed by Prof. Gal Kaminka. I worked on the problem of robotic adversarial coverage, in which a group of robots need to visit every point in a target area, while avoiding threats that might harm them. During my studies I was also a teaching assistant in the courses: Introduction to Robotics and Multi-Robot Teamwork.

July 2015–September 2015: **Research Scientist Intern**, Amazon Prime Air, Israel.

I developed parts of the physical simulator of Amazon's UAV (Unmanned Aerial Vehicle), whose goal is to deliver packages from Amazon's warehouses to customer houses.

July 2008–September 2017: **Lecturer**, The Open University of Israel.

I have taught numerous Computer Science undergraduate and graduate courses, including Advanced Programming, Algorithms, Data Structures, Software Engineering, Computer Structure, Database Systems, Operating Systems, Introduction to Computability Theory and more.

February 2008–July 2008: **Web Developer**, Tegrity, Israel.

I have participated in developing the web site that enables the students to select their desired recordings from their classes and watch them online using a specialized video player.

January 2006–January 2008: **Algorithmic Trading Developer**, ITG, Israel.

I worked as a software developer in the algorithmic trading development team located in the Israeli office, which is responsible for all the European trading activities of the firm. I have quickly become a technical lead in my team, responsible for release of new versions, collaboration with US and Asia-Pacific development teams and system-wide integration.

July 2004–March 2005: **Software Developer**, Motorola, Israel.

I developed a new setup and configuration tool for an instrument called SDM 3000, which gave network fault management solution to ASTRO systems.

December 2001–June 2004: **Senior Software Developer**, Phoenix Insurance Company, Israel.

I developed parts of an insurance system, which enabled insurance agents to create different types of insurance policies, and generate various reports and letters.

February 1996–December 2001: **Senior Software Developer and Team Leader**, The Israeli Army's Intelligence Force.

Education

2013–2018: **Ph.D. in Computer Science**, Bar-Ilan University, Israel
Thesis Title: Robotic Adversarial Coverage
Advisors: Prof. Gal A. Kaminka and Dr. Noa Agmon

In my PhD thesis, I worked on the problem of robotic adversarial coverage, in which a group of robots must visit every point in a target area while avoiding threats that might harm them. The research included both theoretical analysis and practical implementation on physical robot platforms. Portions of this work were published in leading venues in AI and robotics.

2007–2010: **M.Sc. in Computer Science** (Summa cum Laude), The Open University, Israel
Thesis Title: Analysis of the Effects of Lifetime Learning on Population Fitness using Vose Model
Advisors: Prof. Ron Unger and Dr. Mireille Avigal

1997–2001: **B.A. in Computer Science** (Summa cum Laude), The Open University, Israel

Publications

Books

1. Roi Yehoshua. Machine Learning Foundations: Volume 1: Supervised Learning. *Pearson*, forthcoming January 2026. ISBN 978-0135337868.

Journals

1. Roi Yehoshua and Noa Agmon. Capturing an Area-Covering Robot. *Autonomous Agents and Multi-Agent Systems*, 32(3): 313-348, 2018.
2. Roi Yehoshua, Noa Agmon, and Gal A. Kaminka. Robotic Adversarial Coverage of Known Environments. *International Journal of Robotics Research*, 35(12): 1419-1444, 2016.

Highly Refereed Conferences

1. Roi Yehoshua and Christopher Amato. Hybrid Independent Learning in Cooperative Markov Games. In *Proceedings of the Second International Conference on Distributed Artificial Intelligence* (DAI-20), pages 69-84, 2020.
2. Roi Yehoshua and Noa Agmon. Multi-Robot Adversarial Coverage. In *Proceedings of the European Conference on Artificial Intelligence (ECAI-16)*, pages 1493-1501, 2016.
3. Roi Yehoshua and Noa Agmon. Online Robotic Adversarial Coverage. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS-15)*, pages 3830-3835, 2015.
4. Roi Yehoshua, Noa Agmon, and Gal A. Kaminka. Frontier-Based RTDP: A New Approach to Solving the Robotic Adversarial Coverage Problem. In *Proceedings of the 14th International Joint Conference on Autonomous Agents and Multi-Agent Systems (AAMAS-15)*, pages 861-869, 2015.
5. Roi Yehoshua and Noa Agmon. Adversarial Modeling in the Robotic Coverage Problem. In *Proceedings of the 14th International Joint Conference on Autonomous Agents and Multi-Agent Systems (AAMAS-15)*, pages 891-899, 2015.
6. Roi Yehoshua, Noa Agmon, and Gal A. Kaminka. Safest Path Adversarial Coverage. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS-14)*, pages 3027-3032, 2014.
7. Roi Yehoshua, Noa Agmon, and Gal A. Kaminka. Robotic Adversarial Coverage: Introduction and Preliminary Results. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS-13)*, pages 6000-6005, 2013.
8. Roi Yehoshua, Mireille Avigal, and Ron Unger. Analysis of the Effects of Lifetime Learning on Population Fitness using Vose Model. In *Proceedings of the 12th Annual Conference on Genetic and Evolutionary Computation (GECCO-10)*, pages 681-688, 2010.

Abstracts and Short Papers

1. Roi Yehoshua. Robotic Adversarial Coverage (Doctoral Consortium). In *Proceedings of the 14th International Joint Conference on Autonomous Agents and Multi-Agent Systems (AAMAS-15)*, pages 1969-1970, 2015.
   Also presented in the Doctoral Consortium of ICAPS-15.
2. Roi Yehoshua, Noa Agmon, and Gal A. Kaminka. Towards Safest Path Adversarial Coverage (Extended Abstract). In *Proceedings of the 13th International Joint Conference on Autonomous Agents and Multi-Agent Systems (AAMAS-14)*, pages 1383-1384, 2014.

Preprints

1. Roi Yehoshua, Juan Heredia-Juesas, Yushu Wu, Christopher Amato, Jose Martinez-Lorenzo. Decentralized Reinforcement Learning for Multi-Target Search and Detection by a Team of Drones. preprint arXiv:2103.09520, 2021.

Workshops

1. Roi Yehoshua and Noa Agmon. Robotic Adversarial Coverage: From Theory to Practice. In *IJCAI Workshop on Autonomous Mobile Service Robots*, 2016.
2. Roi Yehoshua and Noa Agmon. Multi-Robot Adversarial Coverage. In *IJCAI Workshop on Multi-Agent Path Finding*, 2016.
3. Roi Yehoshua, Noa Agmon, and Gal A. Kaminka. Frontier-Based RTDP: A New Approach to Solving the Robotic Adversarial Coverage Problem. In *ICAPS Workshop on Planning and Robotics, PlanRob-15*, 2015.
4. Roi Yehoshua, Noa Agmon, and Gal A. Kaminka. Towards Efficient Robot Adversarial Coverage. In *AAAI Workshop on Intelligent Robotic Systems*, 2013.
(also presented in BISFAI 2013)

Teaching

2024–2026: **Large Language Models (DS 5983, EECE 5668)**, Northeastern University. This is a new course in the ECE department that I have developed from scratch. The course introduces students to the foundations, methodologies, and implications of large language models. It teaches them how to implement, train, and fine-tune LLMs for various NLP tasks, such as text generation, summarization, and question answering.

2021–2026: **Supervised Machine Learning (DS 5220, EECE 5644)**, Northeastern University. This course covers the theory and practical algorithms for supervised machine learning, including linear regression, K-nearest neighbors, support vector machines, decision trees and neural networks, as well as learning theory (bias/variance tradeoffs, VC theory).

2020–2023: **Unsupervised Machine Learning and Data Mining (DS 5230)**, Northeastern University. This course introduces unsupervised machine learning and data mining, and offers a broad view of models and algorithms for discovering and summarizing patterns from large amounts of unlabeled data.

2021–2024: **Introduction to Data Management and Processing (DS 5110)**, Northeastern University. This course provides a comprehensive introduction to the design of databases and the use of database management systems for data science.

2021–2026: **Software Engineering (EECE 4520)**, Northeastern University. This course offers an overview of the discipline of software engineering. It covers the software life cycle, various models of the software process (structured and agile), the Unified Modeling Language (UML) as applied to the software life cycle, prototyping, design patterns, software metrics and estimation.

2020–2021: **Computational Methods for Data Analytics (EECE 2300)**, Northeastern University. This course introduces the programming tools, algorithms, and software used in data analytics. It covers core concepts of regression analysis, classification, and decomposition.

Spring 2020: **Artificial Intelligence (CS 4100)**, Northeastern University. This is an introductory course on artificial intelligence, that covers a relatively broad range of topics, including search, planning, reasoning under uncertainty, machine learning (supervised and reinforcement), and deep learning.

2018: **Data Science Bootcamp**: A 6-month intensive program I led and developed at Experis Academy to train graduate students to become data scientists in the Israeli tech industry. The curriculum covered Python programming, data manipulation and visualization with NumPy, Pandas and Matplotlib, relational and NoSQL databases (MongoDB), statistical analysis, machine learning (classification, regression, clustering, reinforcement learning), deep learning (CNNs, RNNs, TensorFlow), natural language processing, cloud computing (AWS), big data tools (Spark, Hadoop), and a capstone project based on real-world datasets.

2016–2017: **Advanced Programming**, Bar-Ilan University. This course teaches advanced concepts in programming, such as multi-threaded applications, memory management, socket programming, design patterns, and developing GUI applications.

2014–2016: **Multi-Robot Systems**, Bar-Ilan University. This is an advanced course focusing on algorithms, data structures and system architectures for building multi-robot systems.

2013–2017: **Introduction to Robotics**, Bar-Ilan University. This is an undergraduate level course introducing robot control paradigms, algorithms for common tasks and robot architectures. As a TA in the course, I have prepared a set of tutorials and hands-on labs for learning ROS (Robot Operating System), which have been used in various academic courses worldwide.

2013–2017: **Various courses in Computer Science**, College of Management Academic Studies, Israel. I have taught numerous courses in Computer Science for undergraduates, including: Data Structures, Operating Systems, Database Systems, Introduction to Computation, Advanced Software Development, Cloud Computing, Developing Web Applications, Introduction to Robotics, and Computers Structure.

Book Authoring

I am currently engaged in authoring a comprehensive textbook on Machine Learning, commissioned by Pearson/Addison-Wesley. The textbook covers all aspects of machine learning models, including their theory and mathematical underpinnings, implementation in code, and practical considerations. It consists of three volumes:

1. **Volume I: Supervised Learning**. This volume introduces the core concepts of machine learning, including generalization, model capacity, the bias–variance tradeoff, regularization, and the typical workflow for building and evaluating models. It also covers foundational supervised learning algorithms such as linear and logistic regression, k-nearest neighbors, naive Bayes, decision trees, ensemble methods, and support vector machines, while discussing their assumptions, mathematical foundations, training procedures, and practical applications.

2. **Volume II: Unsupervised and Deep Learning**. This volume explores learning from unlabeled data, including topics such as clustering, dimensionality reduction, density estimation, anomaly detection, and semi-supervised learning. It also provides a rigorous treatment of deep learning, covering various architectures such as feedforward neural networks, convolutional neural networks (CNNs), recurrent neural networks (RNNs), autoencoders, transformers, and graph neural networks (GNNs). The volume combines theoretical depth with practical insights, showcasing applications in diverse areas such as computer vision, medical diagnosis, natural language processing, and recommender systems.

3. **Volume III: Advanced Topics**. This volume covers advanced and emerging areas in machine learning, including deep reinforcement learning, generative AI, and large language models (LLMs). It also delves into foundational principles from statistical learning theory, offering deeper insights into the theoretical guarantees and limitations of learning algorithms. Additional chapters address practical and ethical aspects of machine learning, including model deployment and serving, interpretability, bias and fairness, and adversarial robustness. The volume concludes with a comprehensive survey of current research directions and open questions, preparing the reader for independent investigation and further exploration of this vast field.

Scientific Community Participation

PC Member: AAMAS 2026, SAC 2026, ECAI 2023, AAMAS 2017–2020, ICAPS 2020, AAAI 2019, IJCAI 2019, ECAI 2016, ARMS Workshop in AAMAS 2014–2017.

Reviewer (conferences): AAMAS 2023, ICRA 2020–2023, IROS 2021, ICAPS 2019, IROS 2015–2018, IJCAI 2016.

Reviewer (journals): IEEE RA-L, JMLR, JAAMAS, IEEE Transactions on Robotics, IEEE Intelligent Systems, Autonomous Robots.

Scholarships

- 2013–2017: The Presidential Scholarship for Doctoral Studies in Bar-Ilan
- 2013: Bar-Ilan University outstanding excellence scholarship for Ph.D. students
- 2008: The Open University outstanding excellence scholarship for M.Sc. students
- 1999–2001: The Open University outstanding excellence scholarship for B.A. students